

Ipratropium (oral inhalation): Drug information - UpToDate

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For additional information see "Ipratropium (oral inhalation): Patient drug information" and "Ipratropium (oral inhalation): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Atrovent HFA

Brand Names: Canada

- AA-Ipravent;
- Atrovent HFA;
- JAMP-Ipratropium HFA;
- PMS-Ipratropium;
- TEVA-Ipratropium Bromide

Pharmacologic Category

- Anticholinergic Agent

Dosing: Adult

Asthma, acute exacerbation, moderate to severe

Asthma, acute exacerbation, moderate to severe (off-label use): Note: May consider for treatment of moderate to severe exacerbations (eg, critically ill) in combination with a short-acting beta-adrenergic agonist. Nebulized therapy may be preferred in patients who have more severe symptoms or who cannot effectively use an inhaler (Ref).

Nebulization solution: **Oral inhalation:** 0.5 mg every 20 minutes for 3 doses, then hourly as needed for up to 3 hours (Ref).

Metered-dose inhaler (17 mcg/actuation): **Oral inhalation:** 4 to 8 inhalations with spacer every 20 minutes for 3 doses, then hourly as needed for up to 3 hours (Ref).

Chronic obstructive pulmonary disease

Chronic obstructive pulmonary disease:

Acute exacerbation: Note: Although similar efficacy exists among formulations, some experts prefer nebulized therapy during severe chronic obstructive pulmonary disease (COPD) exacerbations

(Ref). Not recommended as monotherapy, unless patients are intolerant of short-acting beta-agonists (Ref).

For patients at home or office management:

Metered-dose inhaler (17 mcg/actuation): **Oral inhalation:** 2 inhalations every 4 to 6 hours as needed; maximum dose: 12 inhalations per first 12 hours (Ref).

Nebulization solution: **Oral inhalation:** 0.5 mg every 6 to 8 hours as needed; maximum dose: 6 doses (3 mg) per first 12 hours (Ref).

For patients requiring emergency department or hospital-based care:

Metered-dose inhaler (17 mcg/actuation): **Oral inhalation:** 2 to 4 inhalations every 1 to 2 hours (up to 3 doses), then every 2 to 4 hours as needed; maximum dose: 12 inhalations per first 12 hours (Ref).

Nebulization solution: **Oral inhalation:** 0.5 mg every 1 hour (up to 3 doses), then every 2 to 4 hours as needed; maximum dose: 6 doses (3 mg) per first 12 hours (Ref).

Intermittent symptom relief: Note: Use for intermittent symptoms on an as-needed basis rather than regularly scheduled maintenance therapy. Typically used in combination with inhaled short-acting beta agonist (SABA); combination therapy provides greater improvement in FEV₁ and symptoms over monotherapy SABA (Ref).

Metered-dose inhaler (17 mcg/actuation): **Oral inhalation:** 2 inhalations 4 to 6 times daily as needed (Ref).

Nebulization solution: **Oral inhalation:** 0.5 mg every 4 to 8 hours as needed (Ref).

Maintenance (alternative agent): Note: Depending on symptoms and exacerbation risk, preferred therapy is a single long-acting bronchodilator (long-acting beta agonist or long-acting muscarinic antagonist). In patients with more symptoms (eg, Group B), preferred therapy is dual long-acting bronchodilator therapy (Ref).

Metered-dose inhaler (17 mcg/actuation): **Oral inhalation:** 2 inhalations every 6 hours.

Nebulization solution: **Oral inhalation:** 0.5 mg every 6 to 8 hours.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely as limited systemic absorption is expected.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Ipratropium (oral inhalation): Pediatric drug information")

Dosage guidance:

Clinical considerations: A valved holding chamber or spacer is recommended for use with a metered-dose inhaler (MDI) in patients with poor technique and is the recommended medication delivery system in all patients ≤5 years of age, with the addition of a mask for patients <3 or 4 years of age or until proper technique is demonstrated (Ref).

Asthma, acute exacerbation

Asthma, acute exacerbation: Limited data available.

Note: Ipratropium may be added to short-acting beta-2 agonist (SABA) for moderate to severe exacerbations. It may also be added if poor response to SABA therapy during initial management in an acute care setting (eg, emergency department). Ipratropium has not been shown to provide further benefit (eg, after first 24 hours) once the patient is hospitalized (Ref).

Children:

Nebulization: Oral inhalation: 0.25 to 0.5 mg (250 to 500 mcg) every 20 minutes for 1 hour (ie, 3 doses) (Ref).

Metered-dose inhaler: Oral inhalation: 4 to 8 inhalations every 20 minutes as needed for up to 3 doses (Ref).

Adolescents:

Nebulization: Oral inhalation: 0.5 mg (500 mcg) every 20 minutes for 3 doses (Ref).

Metered-dose inhaler: Oral inhalation: 4 to 8 inhalations every 20 minutes as needed for up to 3 doses (Ref).

Bronchospasm associated with chronic pulmonary conditions

Bronchospasm associated with chronic pulmonary conditions: Children ≥ 12 years and Adolescents: Nebulization: 0.5 mg (500 mcg, one unit-dose vial) 3 to 4 times daily with doses 6 to 8 hours apart.

Bronchospasm, wheezing

Bronchospasm, wheezing: Limited data available; efficacy results variable: Infants: Nebulization: 0.125 to 0.25 mg (125 to 250 mcg) every 4 hours has been found helpful in some infants with chronic or recurrent wheezing, and some patients with bronchiolitis; however, most bronchiolitis data suggests ipratropium is not effective (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling (not studied); however, dosage adjustment unlikely necessary due to low systemic absorption.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling (not studied); however, dosage adjustment unlikely necessary due to low systemic absorption.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Respiratory: Bronchitis (10% to 23%), exacerbation of chronic obstructive pulmonary disease (8% to 23%), sinusitis (1% to 11%)

1% to 10%:

Central nervous system: Headache (6% to 7%), dizziness (3%)

Gastrointestinal: Dyspepsia (1% to 5%), nausea (4%), xerostomia (2% to 4%), dysgeusia (1%)

Genitourinary: Urinary tract infection (2% to 10%)

Neuromuscular & skeletal: Back pain (2% to 7%)

Respiratory: Dyspnea (7% to 8%), flu-like symptoms (4% to 8%), cough (>3%), rhinitis (>3%), upper respiratory tract infection (>3%)

<1%, postmarketing, and/or case reports: Accommodation disturbance, acute eye pain, anaphylaxis, angioedema, blurred vision, bronchospasm, conjunctival hyperemia, constipation, corneal edema,

decreased gastrointestinal motility, diarrhea, dry throat, glaucoma, hypersensitivity reaction, hypotension, increased intraocular pressure, laryngospasm, mouth edema, mydriasis, nausea, palpitations, pharyngeal edema, pruritus, skin rash, stomatitis, tachycardia, throat irritation, urinary retention, urticaria, visual halos around lights, vomiting

Contraindications

Hypersensitivity to ipratropium, atropine (and its derivatives), or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

- **Bronchospasm:** Paradoxical bronchospasm that may be life-threatening and may occur with use of inhaled bronchodilating agents; this should be distinguished from inadequate response. If paradoxical bronchospasm occurs, discontinue ipratropium and institute alternative therapy.
- **CNS effects:** May cause dizziness and blurred vision; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving).
- **Hypersensitivity reactions:** Hypersensitivity reactions (urticaria, angioedema, rash, bronchospasm, oropharyngeal edema), including anaphylaxis, have been reported. Discontinue therapy immediately if patient develops an allergic reaction.

Disease-related concerns:

- **Glaucoma:** Use with caution in patients with narrow-angle glaucoma; may increase intraocular pressure.
- **Prostatic hyperplasia/bladder neck obstruction:** Use with caution in patients with prostatic hyperplasia or bladder neck obstruction; may cause urinary retention.

Other warnings/precautions:

- **Appropriate use:** Not indicated for the initial treatment of acute episodes of bronchospasm where rescue therapy is required for rapid response. Only use in acute exacerbations of asthma in conjunction with short-acting beta-adrenergic agonists for acute episodes (NAEPP 2007).

Dosage Forms Considerations

Atrovent HFA 12.9 g canister contains 200 inhalations.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Aerosol Solution, Inhalation, as bromide:

Atrovent HFA: 17 mcg/actuation (12.9 g)

Generic: 17 mcg/actuation (12.9 g)

Solution, Inhalation, as bromide:

Generic: 0.02% (2.5 mL)

Solution, Inhalation, as bromide [preservative free]:

Generic: 0.02% (2.5 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Aerosol solution (Atrovent HFA Inhalation)

17 mcg/ACT (per gram): \$28.54

Aerosol solution (Ipratropium Bromide HFA Inhalation)

17 mcg/ACT (per gram): \$28.47

Solution (Ipratropium Bromide Inhalation)

0.02% (per mL): \$0.14 - \$1.44

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Aerosol Solution, Inhalation:

Atrovent HFA: 20 mcg/actuation (1 ea) [contains alcohol, usp]

Generic: 20 mcg/actuation (10 mL)

Nebulization Solution, Inhalation:

Generic: 0.125 mg/mL (2 mL); 0.25 mg/mL (0.5 mL, 1 mL, 2 mL, 20 mL)

Administration: Adult

For oral inhalation; avoid spraying in eyes.

Metered-dose inhaler: Prior to initial use, prime inhaler by releasing 2 test sprays into the air; inhaler does not require shaking. If the inhaler has not been used for >3 days, reprime. Wash mouthpiece once a week for 30 seconds in warm water only and let air dry. Discard inhaler once dose indicator displays "0."

Nebulization solution: Remove unit dose vial from foil pouch and squeeze contents into nebulizer reservoir; connect nebulizer to compressor. Breathe deeply and evenly until all medication has been inhaled; inhalation time is about 5 to 15 minutes. Clean nebulizer after use. Compatibility with other medications (eg, albuterol, budesonide) in nebulizer has been reported (Ref); also refer to institution-specific policies.

Administration: Pediatric

Inhalation:

Nebulization: May be administered with or without dilution in NS. Blow-by administration is not recommended; use of a nebulizer with a mouthpiece, rather than a face mask, may be preferred to prevent contact with eyes; a facemask should be used if patient unable to hold mouthpiece in mouth for administration or if <4 years of age (Ref). Compatibility with other medications (eg, albuterol, budesonide) in nebulizer has been reported (Ref); also refer to institution-specific policies.

Oral Inhalation (metered-dose inhaler [MDI]): Atrovent HFA: Prior to initial use, prime inhaler by releasing 2 test sprays into the air. If the inhaler has not been used for >3 days, reprime. Avoid spraying into the eyes. A valved holding chamber or spacer is recommended for use with an MDI in patients with poor technique and is the recommended medication delivery system in all patients ≤5 years of age, with the addition of a mask for patients <3 or 4 years of age or until proper technique is demonstrated (Ref).

Storage/Stability

Metered-dose inhaler: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Exposure to temperatures >48°C (120°F) may cause bursting. Do not puncture inhaler, throw into a fire or incinerator, or use or store near heat or open flame.

Nebulization solution: Store at 15°C to 30°C (59°F to 86°F). Protect from light. Store unused vials in foil pouch.

Use: Labeled Indications

Chronic obstructive pulmonary disease: Maintenance treatment of bronchospasm associated with chronic obstructive pulmonary disease (COPD), including chronic bronchitis and emphysema

Use: Off-Label: Adult

Asthma, acute exacerbation, moderate to severe

Medication Safety Issues

Sound-alike/look-alike issues:

Atrovent may be confused with Alupent, Serevent

Ipratropium may be confused with tiotropium

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: Ipratropium (Oral Inhalation) may increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Inhaled Anticholinergic Agents: May increase anticholinergic effects of Inhaled Anticholinergic Agents. *Risk C: Monitor*

Loxapine: Agents to Treat Airway Disease may increase adverse/toxic effects of Loxapine. More specifically, the use of Agents to Treat Airway Disease is likely a marker of patients who are likely at a greater risk for experiencing significant bronchospasm from use of inhaled loxapine. Management: This is specific to the Adasuve brand of loxapine, which is an inhaled formulation. This does not apply to non-inhaled formulations of loxapine. *Risk X: Avoid*

Methacholine: Ipratropium (Oral Inhalation) may decrease therapeutic effects of Methacholine. Management: Hold ipratropium for 12 hours before methacholine use. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Systemic exposure following inhalation is negligible; maternal use is not expected to result in fetal exposure. Data related to ipratropium for the treatment of asthma in pregnancy are limited. Based on available information, an increased risk of adverse maternal or fetal outcomes has not been observed following use during pregnancy.

Uncontrolled asthma is associated with adverse events on pregnancy (increased risk of perinatal mortality, preeclampsia, preterm birth, low birth weight infants, cesarean delivery, and the development of gestational diabetes). Poorly controlled asthma or asthma exacerbations may have a greater

fetal/maternal risk than what is associated with appropriately used asthma medications. Maternal treatment improves pregnancy outcomes by reducing the risk of some adverse events (eg, preterm birth, gestational diabetes). Maternal asthma symptoms should be monitored every 4 to 6 weeks during pregnancy (ERS/TSANZ [Middleton 2020]; GINA 2024).

Breastfeeding Considerations

It is not known if ipratropium is present in breast milk.

Systemic exposure following inhalation is negligible which would limit excretion into breast milk. According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

FEV₁, peak flow, and/or other pulmonary function tests; signs/symptoms of glaucoma; hypersensitivity reactions; urinary retention

Mechanism of Action

Blocks the action of acetylcholine at parasympathetic sites in bronchial smooth muscle causing bronchodilation; local application to nasal mucosa inhibits serous and seromucous gland secretions.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Bronchodilation: Within 15 minutes

Peak effect: 1 to 2 hours

Duration: Metered-dose inhaler: 2 to 4 hours; Nebulization solution: 4 to 5 hours, up to 7 to 8 hours in some patients

Absorption: Not readily absorbed into the systemic circulation from the surface of the lung or from the GI tract; ~7% absorbed after nebulization of a 2 mg dose

Distribution: 15% of dose reaches lower airways

Protein Binding: ≤9%

Metabolism: Partially metabolized to inactive ester hydrolysis products

Half-life elimination: 2 hours

Excretion: Urine (50%)

Patient Counseling Points

(For additional information see "Ipratropium (oral inhalation): Patient drug information")

What is this drug used for?

- It is used to open the airways in lung diseases where spasm may cause breathing problems.
- This drug is not to be used to treat intense flare-ups of shortness of breath. Use a rescue inhaler. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Back pain
- Headache
- Dry mouth
- Flu-like symptoms
- Nasal irritation

- Sore throat
- Stuffy nose

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Urinary tract infection like blood in the urine, burning or painful urination, passing a lot of urine, fever, lower abdominal pain, or pelvic pain.
- Vision changes
- Eye pain
- Severe eye irritation
- Seeing halos or bright colors around lights
- Red eyes
- Trouble urinating
- Unable to pass urine
- Passing a lot of urine
- Severe dizziness
- Passing out
- Fast heartbeat
- Abnormal heartbeat
- Mouth sores
- Mouth irritation
- Trouble breathing
- Wheezing
- Cough
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

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Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Atrovent;
- (AR) Argentina: Aerotrop | Atrovent | Atrovent hfa | Iprabron;
- (AT) Austria: Atrovent;
- (AU) Australia: Atrovent;
- (BD) Bangladesh: Ipramid | Iprex;
- (BE) Belgium: Atronase | Atrovent;
- (BG) Bulgaria: Atrovent | Atrovent n;
- (BR) Brazil: Atrovent | Atrovent n;
- (CH) Switzerland: Atrovent n;
- (CL) Chile: Aerotrop | Atrovent | Influmix | Iprasynt hfa;
- (CN) China: Atrovent;
- (CO) Colombia: Aspromio | Atrovent | Atrovent hfa | Biovent | Bromuro de ipratropio | Ciplatropium | Ipramar | Iprasynt | Ipratropio bromuro | Plamune;
- (CR) Costa Rica: Aerotrop | Atroaldo | Bromuro de ipratropio | Ipratropio bromuro | Ipravent;
- (CZ) Czech Republic: Atrovent | Atrovent n;
- (DE) Germany: Atrovent | Atrovent n | Iprabronch | Ipratropiumbromid hexal | Ipravent;
- (DO) Dominican Republic: Aerotrop | Atrovent | I Patrimul | Ipratropium | Tropium;
- (EC) Ecuador: Aspromio | Atrovent | Atrovent hfa | Bromuro de ipratropio | I Patrimul | Iprasynt | Ipratropio | Ipratropio bromuro | Ipravent | Protaiisol;
- (EE) Estonia: Atrovent | Atrovent n;
- (EG) Egypt: Atrovent;
- (ES) Spain: Atroaldo | Atrovent | Ipratropio bromuro cipla | Ipratropio bromuro Kern Pharma;
- (FI) Finland: Atrovent | Atrovent eco;
- (FR) France: Atrovent;
- (GB) United Kingdom: Atrovent | Inhalvent | Ipratropium | Ipravent | Ipravent cfc;
- (GR) Greece: Atrovent | Broncovent;
- (GT) Guatemala: Aspromio | Ipratropio;
- (HK) Hong Kong: Atrovent | Ipratropium bromide aldo union;
- (HR) Croatia: Atrovent n;
- (HU) Hungary: Atrovent | Atrovent n | Ipravent;
- (ID) Indonesia: Atrovent;
- (IE) Ireland: Atrovent;
- (IL) Israel: Aerovent | Atrovent;
- (IN) India: Ipramist | Ipratrop | Ipravent;
- (IQ) Iraq: Ipravent;
- (IT) Italy: Atem | Rinovagos;
- (JO) Jordan: Atrovent;
- (JP) Japan: Atrovent;
- (KE) Kenya: Atrovent hfa;
- (KR) Korea, Republic of: Atrovent;
- (KW) Kuwait: Atrovent | Atrovent n;
- (LB) Lebanon: Atem | Atrovent | Atrovent n;
- (LT) Lithuania: Atrovent | Atrovent n;
- (LU) Luxembourg: Atronase | Atrovent;
- (LV) Latvia: Atrovent;
- (MX) Mexico: Atrovent | Bromuro de ipratropio | Ipralong | Iprasynt | Ipratropio | Protaiisol;
- (MY) Malaysia: Atrovent | Ipracid | Ipravent;
- (NL) Netherlands: Atrovent | Atrovent(es) | Ipratropiumbromide Sandoz;
- (NO) Norway: Atrovent;
- (NZ) New Zealand: Atrovent;
- (PA) Panama: Aerotrop | Aspromio | Atroaldo | Atrovent | Bromuro de ipratropio | Protaiisol;
- (PE) Peru: Aerotrop | Atrovent | Atrovent hfa | Bromuro de ipratropio | Flavuril | Ipramar | Iprapharm | Ipratrolab | Ipratropium;

- (PH) Philippines: Atrovent;
- (PK) Pakistan: Atem | Estravent | Ipravent | Optra | Optra HFA | Trupium;
- (PL) Poland: Atrodil | Atrovent | Atrovent n | Ipravent;
- (PR) Puerto Rico: Atrovent | Atrovent hfa | Ipratropium br;
- (PT) Portugal: Atrovent | Atrovent pa | Brometo de ipratropio bluepharma;
- (PY) Paraguay: Aerotrop | Anasmal | Atrovent hfa | Bromovent;
- (QA) Qatar: Atrovent MDI | Atrovent UDV | Ipravent;
- (RO) Romania: Atrovent | Ipravent;
- (RU) Russian Federation: Atrovent | Atrovent n | Ipratropium aeronativ | Ipratropem | Ipravent;
- (SA) Saudi Arabia: Atrovent;
- (SE) Sweden: Atrovent | Ipratropiumbromid Ebb | Ipravent;
- (SG) Singapore: Atrovent | Iprovent;
- (SI) Slovenia: Atrovent;
- (SK) Slovakia: Atrovent;
- (SR) Suriname: Atrovent | Ipravent | Iprex;
- (TH) Thailand: Atrovent;
- (TN) Tunisia: Atroaldo | Atrovent;
- (TR) Turkey: Atrovent | Ipracort;
- (TW) Taiwan: Atrovent;
- (UA) Ukraine: Atrovent | Ipratropium inteli | Ipravent;
- (UY) Uruguay: Aerotrop | Atrovent hfa | Estravent;
- (VE) Venezuela, Bolivarian Republic of: Alovent | Atroaldo | Bromuro de ipratropio | Ipatrixair | Iprasynt | Ipratropio bromuro;
- (ZA) South Africa: Atronase | Atrovent | Ipvent

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